

Name: _____

These questions assess your understanding of the material from Chapters 20, 21 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units.

Note that I provide references on practice exams so you can find additional help if you cannot solve a problem; I do not give this information on in-class exams. The related chapter and section are provided after each question. E.g. "(Chapter 18, Section (5) Electric Field Lines - cp1e_ex18_4)" means that the question was inspired by OpenStax College Physics First Edition's Chapter 18, Example 4, which is found within Chapter 18, Section 5. See the end of this practice exam for a list of the sources.

1. While starting an engine, a truck battery sets in motion $1.1 \times 10^3 \text{ C}$ of charge in 1.4 s. Calculate the current involved in starting the engine.
(Chapter 20, Section (1) Current - cp1e_ex20_1)

ANSWER: _____

2. Solve for the power dissipated in resistor R_6 given $V_1 = 10 \text{ V}$, $V_2 = 10 \text{ V}$, $V_3 = 6 \text{ V}$, $V_4 = 5 \text{ V}$, $R_1 = 1 \Omega$, $R_2 = 5 \Omega$, $R_3 = 2 \Omega$, $R_4 = 4 \Omega$, $R_5 = 2 \Omega$, and $R_6 = 4 \Omega$.
(Chapter 21, Section (3) Kirchhoff's Rules - tkd_circ2)

ANSWER: _____

3. Suppose the voltage supplied to a circuit is 9.6 V. There are three resistors wired in series: $R_1 = 5.0 \Omega$, $R_2 = 3.2 \Omega$, and $R_3 = 3.8 \Omega$. Calculate the power output of the battery.
(Chapter 21, Section (1) Resistors in Series and Parallel - cp1e_ex21_2)

ANSWER: _____

4. A circuit has two branches connected in parallel: branch A consists of resistors $R_1 = 1.1 \Omega$ and $R_2 = 8.0 \Omega$ in series; branch B consists of resistors $R_3 = 4.1 \Omega$ and $R_4 = 1.3 \Omega$ in parallel. Draw this circuit and find the total equivalent resistance. Note: you must draw the circuit correctly to receive credit.
(Chapter 21, Section (1) Resistors in Series and Parallel - cp1e_ex21_3)

ANSWER: _____

5. A car headlight filament is made of tungsten ($\rho = 5.6 \times 10^{-8} \Omega \cdot \text{m}$) and has a resistance of 0.45Ω . If the filament is a cylinder 1.5-cm long, what is its diameter?
(Chapter 20, Section (3) Resistance and Resistivity - cp1e_ex20_5)

ANSWER: _____

6. What is the resistance of an automobile headlight which carries a current of 1.5 A when 6.4 V is applied to it?
(Chapter 20, Section (2) Ohm's Law - cp1e_ex20_4)

ANSWER: _____

7. What current flows through a high-voltage transmission line to transmit 91. MW of power at 140 kV?
(Chapter 20, Section (5) Alternating Current v. Direct Current - cp1e_ex20_10)

ANSWER: _____

8. A 12.8-V car battery with internal resistance $0.018\ \Omega$ supplies a starter motor drawing 510 A for 3.2 s. Calculate the electrical energy dissipated as heat in the battery's internal resistance during this time.
(Chapter 21, Section (2) Electromotive Force: Terminal Voltage - cp1e_ex21_4)

ANSWER: _____

9. What is the value of the peak voltage for 110-V AC power?
(Chapter 20, Section (5) Alternating Current v. Direct Current - cp1e_ex20_9)

ANSWER: _____

10. A 11.3-V car battery supplies a starter motor drawing 440 A for 2.0 s. Calculate the total charge moved by the battery during this time.
(Chapter 20, Section (1) Current - tkd0)

ANSWER: _____

11. A 12.7-V car battery supplies a starter motor drawing 200 A for 1.3 s. Calculate the electrical energy delivered by the battery during this time.
(Chapter 20, Section (4) Electric Power and Energy - tkd1)

ANSWER: _____

12. Suppose the voltage supplied to a circuit is 4.2 V. There are three resistors wired in parallel: $R_1 = 9.7\ \Omega$, $R_2 = 4.3\ \Omega$, and $R_3 = 1.6\ \Omega$. Calculate the current leaving the battery.
(Chapter 21, Section (1) Resistors in Series and Parallel - cp1e_ex21_1)

ANSWER: _____

13. Calculate the power dissipated by a car headlight that draws a current of 5.8 A when it has a voltage applied to it of 15. V.
(Chapter 20, Section (4) Electric Power and Energy - cp1e_ex20_7)

ANSWER: _____

14. An incandescent lightbulb has a 2.4-cm-long filament and operates at 106. V to deliver 27. W. Calculate the diameter of the bulb's filament, assuming it to be a cylindrical tungsten wire, given that the resistivity of tungsten is $5.6 \times 10^{-8}\ \Omega \cdot \text{m}$.
(Chapter 20, Section (4) Electric Power and Energy - tkd2)

ANSWER: _____

15. A 36.-m-long copper wire must carry 19. A without dissipating more than 28. W as heat. Calculate the wire's minimum diameter given that copper's resistivity is $1.7 \times 10^{-8}\ \Omega \cdot \text{m}$.
(Chapter 20, Section (4) Electric Power and Energy - tkd4)

ANSWER: _____

16. Solve for the voltage drop across resistor R_5 given $V_1 = 10\ \text{V}$, $V_2 = 6\ \text{V}$, $V_3 = 6\ \text{V}$, $R_1 = 4\ \Omega$, $R_2 = 1\ \Omega$, $R_3 = 2\ \Omega$, $R_4 = 4\ \Omega$, and $R_5 = 4\ \Omega$.
(Chapter 21, Section (3) Kirchhoff's Rules - tkd_circ1)

ANSWER: _____

17. A 2.0-m-long copper wire of diameter 1.1 mm is connected to a 7.6-V battery at 3.5 °C. Calculate the percent change in the power delivered by the wire when it is heated to 73. °C. The resistivity of copper at 3.5 °C is $1.7 \times 10^{-8} \Omega \cdot \text{m}$ and its linear thermal expansion coefficient is $3.9 \times 10^{-3} \frac{1}{^\circ\text{C}}$.

(Chapter 20, Section (4) Electric Power and Energy - tkd3)

ANSWER: _____

cp1e: College Physics (1st Edition) by OpenStax

PSE_Serway_4e: Physics for Scientists and Engineers (4th Edition) by Serway

tkd: I did not refer to anything while writing this question